

Moaz Hany El Mahdy

AI/ML Engineer

✉ moazhany27@gmail.com ☎ +201557992912 📍 Cairo, Egypt 🇪🇬 Egyptian  Moaz Hany

🔗 Eng MoazHany

Professional Summary

AI/ML Engineer with hands-on experience building machine learning solutions, FastAPI microservices, and production-ready web applications. Experienced in forecasting models, REST APIs, AI integration, and data-driven applications. Professional experience at Norm Production and a graduate of Capital University (2026). Skilled in Python, machine learning, backend development, and AI system integration, with continuous growth in Deep Learning, NLP, and MLOps through the DEPI Microsoft ML Engineer Track.

Education

Bachelor of Computer Science & Artificial Intelligence (Information Systems), 2026
Capital University (formerly Helwan University) Cairo, Egypt

Technical Skills

Machine Learning & AI:

- Supervised Learning, Regression, Classification, Feature Engineering, Data Preprocessing, Model Evaluation, Scikit-learn

Deep Learning

- Neural Networks (Keras)
- Activation Functions, Loss Functions
- Basics of CNNs & NLP Models

Tools & Platforms

Git, GitHub, Docker, Postman, VS Code

Frontend Development

React

Data & Visualization

- Pandas, NumPy, Matplotlib, Data Cleaning, Data Preprocessing

Databases

Projects

FINEXA – AI-Powered Personal Finance Management Platform,

Graduation Project | AI/ML Engineer & Backend Integration

- Developed and integrated a FastAPI AI microservice for personal finance forecasting and smart saving recommendations.
- Built an expense forecasting pipeline using Python, pandas, NumPy, and scikit-learn.
 - Trained a Random Forest Regression model with lag features, rolling averages, category encoding, and log-target transformation.
- Implemented category normalization and mapping logic while maintaining frontend-compatible categories.
- Designed an explainable rule-based Smart Saving Plan Advisor with financial guardrails and recommendation logic.
- Developed REST APIs for forecasting, saving plans, health checks, and Swagger documentation.
- Integrated the AI service through the main ASP.NET Core backend rather than direct frontend communication.

Student Performance Predictor (ML App)

- Regression & Neural Network model with preprocessing, Early Stopping, LR Scheduler, Dropout, and Streamlit deployment.

Sudoku Solver (GA + Backtracking)

- Hybrid solver using Genetic Algorithm (mutation, crossover, fitness) + classical Backtracking to optimize solving time.

Data Mining Project

- Applied Decision Tree, SVM, and KNN with preprocessing (scaling, encoding) and evaluation via cross-validation.

Certificates

- | | | |
|--|--|--|
| • IBM - Deep Learning with Keras and Tensorflow (2026) | • IBM - Machine Learning with Python (2025) | • IBM - Introduction to Deep Learning & Neural Networks with Keras(2025) |
| • DEPI - Microsoft ML Engineer Track (2025-Present) | • Introduction to Artificial Intelligence & Applications (Zewail City - Certificate of Attendance) | • Time Management & Shopping Behavior (EG Bank - Training Session) |

Languages

- | | |
|----------|-----------|
| • Arabic | • English |
|----------|-----------|